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Evaluating a seasonal autoregressive moving average model with an exogenous variable for short-term timber price forecasting

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ABSTRACT

In this paper roundwood prices were forecasted by means of the following time series models: (1) classical univariate autoregressive moving average (ARIMA), (2) seasonal univariate (SARIMA), and (3) seasonal bivariate with an exogenous variable (SARIMAX). The data consisted of time series of the nominal prices of pine, spruce, beech, birch, and alder roundwood and the construction confidence index (CCI) in Poland for the years 2005–2020. The results indicate that roundwood prices can be evaluated by the implemented time series models, with forecast accuracy depending on the time horizon, the model applied, and tree species. The incorporation of a separate component directly describing seasonal fluctuations into a conventional ARIMA time series model significantly improved the accuracy of roundwood price forecasts exhibiting seasonal variation. The CCI lagged by one to three quarters were found to be significantly correlated with the prices of roundwood. A SARIMAX model incorporating the CCI as a leading variable was able to account for changes in the business cycle, thus increasing the accuracy of short-term forecasts of the prices of roundwood used in the construction industry.

1. Introduction

Forecasting is closely associated with human planning activity as it enables the prediction of upcoming events, facilitates business decisions, and makes it possible to estimate their impact on future economic developments (Johnston and Buongiorno, 2017; Prestemon and Holmes, 2000). The ability to predict market behavior, including the demand and supply of specific goods and the resulting price levels offers a substantial competitive advantage for companies in free markets, especially given the increasing globalization. The rapid development of information technologies and widespread accumulation of data in numerous sectors provides unique opportunities to generate forecasts.

In the forest sector, forecasting typically concerns timber supply, demand, and prices (Yin and Baek, 2004), with the two broad approaches involving structural and nonstructural models. What distinguishes structural models from nonstructural ones, and some structural models from others, is how clearly they make connections between institutional, economic, and statistical assumptions and the estimated relationships. In structural econometric models, economic theory is used to develop mathematical statements about how a set of observable endogenous variables, are related to another set of observable (and sometimes unobservable) exogenous variables (Reiss and Wolak, 2007). Such models usually take the form of a system of simultaneous equations in which each endogenous variable is a function of other endogenous variables as well as the exogenous variable and the error term (Chen et al., 1988; Myneni et al., 1994). Structural models are particularly suitable for making long-term and large-scale forecasts.

In nonstructural time series models there is no a priori distinction into *endo-* and exogenous variables, neither there is a rigorous economic theory that would provide foundations for the model (Charemza and Deadman, 1992). Such models may be based only on the time series of one variable, as in the case of ARIMA (Leskinen and Kangas, 1998) or on the regression of each current variable with respect to all other variables lagged by a certain number of periods, as in the case of VAR (Baek and Yin, 2006; Parajuli and Chang, 2015). These models mostly use disaggregated data and are particularly suitable for short-term forecasting and analysis of separate market segments taking into account different regions, species, or types of assortments (Yin and Baek, 2005).

While timber prices are a key variable in forest management optimization, forecasting thereof is fraught with great uncertainty (Leskinen and Kangas, 2001). Time series methods provide strong results while requiring a modest data input, which makes them uniquely useful in the

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analysis of forest economic issues (Tzanova, 2017). Time series analysis makes it possible to assess the effects of external economic shocks on timber price changes. Yin and Newman (1999) successfully implemented intervention analysis to assess the impact of Hurricane Hugo on regional stumpage prices. Yin (2001) applied intervention analysis and causality test for assessing the impact of federal policy shifts (regarding timber harvest restriction triggered by spotted owl protection) and booming housing market on log price changes in the Pacific Northwest. A better understanding of the time series properties of timber prices is crucial for optimal harvest timing, the rational behavior of timber owners, and gives an insight into the structure of markets (Prestemon, 2003). Depending on the time horizon, forecasts can be divided into short-term (in forestry, generally from a month to a year), medium-term (from over a year to 10 years), and long-term (from over 10 years to more than 100 years).

1.1. Long-term forecasts in the forest sector

The existing literature on forest sector forecasting primarily deals with long-term scenarios. Buongiorno (2015a) developed the Global Forest Products Model which simulates how the global demand and supply of multiple products in many countries determine prices, production, consumption, and trade. For a period of more than fifty years, forecasts of the production and prices of selected timber assortments were made based on the demographic and economic growth predictions of the Intergovernmental Panel on Climate Change. The same author also analyzed the long-term influence of potential CO_2 fertilization on the global forest sector; he found that in the long run it is going to increase the world's forest stock as well as timber supply, which will in turn lead to a modest decrease in the prices of end products and higher global consumption (Buongiorno, 2015b).

Long-term price trends and their effects on harvesting decisions were modeled by Khajuria et al. (2012) as a mean-reverting process with a stochastic trend. The prices forecasted by the model were used in real options analysis to determine optimal investment time and optimal investment rule. The impact of two possible types of changes in the capacity of the pulp and paper industry (a new machine or a close-down) on roundwood prices were analyzed on a regional and national scale by Bolkesjø (2005), who found that the price decrement in the case of a close-down was greater than the price increment in the case of a capacity increase. Lundholm et al. (2019) incorporated dynamic timber prices in a decision support system to reflect the changing demands for wood fiber as part of European Union and global efforts to mitigate climate change over a 100-year planning period; they concluded that wood assortment prices increased due to higher demand and the use of wood fiber with a view to mitigating climate change.

Long-term forecasts are fraught with considerable uncertainty and are characterized by a greater degree of generality, with their objective being the determination of long-term trends and changes in a given variable over a long time horizon. Analyzing the consequences of data and parameter uncertainty for long-term projections in a global forest sector model, Buongiorno and Johnston (2018) found that parameter uncertainty led to uncertainty of the projections increasing steadily with the horizon length, more rapidly than the uncertainty stemming from initial conditions. In Kallio and Maarit (2010) model developed for Finland, the global prices of forest industry products (assumed to be exogenous) were the main source of projection uncertainty.

1.2. Short-term forecasts based on time series models

The world's forest products markets have become increasingly interrelated due to globalization and internationalization and react ever more rapidly to fast-changing market conditions (Hetemäki and Mikkola, 2005). Therefore, short-term analysis and forecasting play an essential role in the forest industry and timber markets. Yin (1999) used univariate ARIMA models, adopting the Box and Jenkins modeling procedure, for evaluating selected timber market in US price forecast and stated that short-term forecasts (and especially one-lead forecasts) are fairly accurate. Hetemäki et al. (2004) successfully used univariate and multivariate time series models for short-term forecasting of the Finnish forest sector, suggesting that a system developed on their basis may serve as a useful tool for short-term timber price predictions. Kolo and Tzanova (2017) used a VECM based on time series data for short- and medium-term forecasting of raw timber export and import volumes as well as prices.

In the forecasting of variables exhibiting seasonal fluctuations, widespread use is made of SARIMA models, which have been applied for predicting demand in the aviation industry (Xu et al., 2019), short-term electricity demand and prices (Zhang et al., 2020), and the prices of agricultural products (Sabu and Kumar, 2020). Univariate time series models based only on an endogenous variable (ARIMA and SARIMA) enable relatively accurate projections as long as the course of the process in the forecast period is similar to that in preceding periods. However, if the variable to be forecasted is highly volatile, projections based on such models will be unsatisfactory. This problem may be addressed by SARIMAX models, which incorporate exogenous leading variables to account for additional information affecting prices not contained in the historical time series data; examples of such variables include those reflecting demand for timber. According to Prestemon et al. (2018), residential construction is a primary user of wood in the United States, and so housing starts have a major impact on the timber market.

The objective of the work was to apply ARIMA, SARIMA and SARIMAX models for short-term timber price forecasting and to evaluate the accuracy of the forecasts generated by these models. The paper is organized as follows. First, the procedures of univariate ARIMA modeling with extension of seasonal fluctuations and exogenous variables are introduced. Next, the data and examples of short-term timber price forecasting are presented, and the accuracy of the forecasts generated by different models are evaluated. The paper ends with concluding remarks.

2. Methods

2.1. Description of the models

Elementary time series models can be divided into autoregressive models (AR) explaining current observations based on the past behavior of a given process, and moving average models (MA), which are linear regressions of the current value of a time series against the values of previous error terms. ARIMA models combining AR and MA are defined as ARIMA (p, d, q), where p and q are the orders of AR and MA models, respectively, and d represents the degree of series integration (differencing) (Yin, 1999):

$$y_t = \varphi_1 y_{t-1} + \varphi_2 y_{t-2} + \ldots + \varphi_p y_{t-p} + e_t + \theta_1 e_{t-1} + \theta_2 e_{t-2} + \ldots + \theta_q e_{t-q} \quad (1)$$

where: y_t is the value observed at time t in the time series, δ is a constant term, φ_i is the i-th autoregressive coefficient, θ_j is the j-th moving average coefficient, and e_t represents residues uncorrelated with any previous ones.

The above model can be expressed in following form:

$$\left(1-\varphi_1 L-\varphi_2 L^2-\ldots-\varphi_2 L^p\right) y_t = \left(1+\theta_1 L+\theta_2 L^2 \ldots+\theta_q L^q\right) e_t \quad (2)$$

where L is the lag operator $L^p y_t = y_{t-p}$.

Seasonal time series can be analyzed by incorporating seasonal fluctuations in classical ARIMA models in the form of parameters (P, D, Q)^S describing seasonal lags. Such models are denoted as SARIMA (p, d, q) (P, D, Q^S), where P – order of seasonal autoregression, D – degree of seasonal integration, Q – order of seasonal moving average, and S – length of seasonal cycle.

In addition to the endogenous variables used in SARIMA, multivariate SARIMAX models incorporate exogenous variables to reflect the

structural changes of the analyzed process.

2.2. Model development steps

Time series modeling requires the system to be stationary. The stationarity of data was evaluated using the augmented Dickey–Fuller (ADF) and Kwiatkowski–Phillips–Schmidt–Shin (KPSS) tests. In the former, the null hypothesis is that the process is not stationary, which is rejected only if evidence to the contrary is strong enough, so it conservatively supports nonstationarity. In contrast, the null hypothesis of the KPSS test states that the time series is nonstationary, and so it conservatively supports data stationarity.

ARIMA model estimation follows the three-stage iterative procedure pioneered by Box and Jenkins (1976): 1) identifying a tentative model, 2) estimating this model, and 3) verifying if the model is adequate:

- Model Identification. The structure and order of models were identified by analyzing the shape of the time series as well as applying an autocorrelation function (ACF) and a partial autocorrelation function (PACF).
- Model estimation. The parameters of tentative models with a high order of lags were estimated and the significance of parameters was evaluated. In subsequent steps, the model was tested down to remove the parameters (lags) that were not statistically significant. The tentative model that best fit the data was identified using the Akaike information criterion (AIC) and the Bayesian information criterion (BIC) according to the following formulas:

$$AIC = 2k + \left[ln\left(\frac{RSS}{n_{obs}}\right) \right], \tag{3}$$

$$BIC = n_{obs}\ ln\left(\sigma^2\right) + k\ ln(n_{obs}) \tag{4}$$

where: k is the number of free parameters, n_{obs} is the total number of observations, RSS is the residual sum of squares, and σ^2 is the error variance. The penalty term in the BIC is larger than in the AIC. Both information criteria represent a trade-off between fit (measured by log likelihood) and parsimony (measured by the number of free parameters). The lower the AIC or BIC, the better the model.

- Diagnostic Checking. For the selected optimal model, goodness of fit was tested by analysis of the estimated model residuals in terms of their normality and autocorrelation (ACF and PACF).

The construction a SARIMAX model involved an additional step of selecting a leading exogenous variable. Candidate variables were tested in terms of whether (1) they were significantly correlated with the endogenous variable (timber prices) by evaluating linear regression parameters, (2) their values were known (available) for the analyzed period of time taking into consideration the lag, (3) they were statistically significant upon incorporation into the SARIMA model (Student's t-test).

Forecast accuracy was evaluated using the root mean square error (RMSR), mean absolute percent error (MAPE), and Theil's inequality coefficient (TIC), according to the formulas given below:

$$RMSR = \sqrt{\frac{1}{n}\sum\nolimits_{t=1}^{n}(\hat{y}_t - y_t)^2} \tag{5}$$

$$MAPE = \frac{1}{n}\sum_{t=1}^{n}\frac{|\hat{y}_t - y_t|}{y_t}\ 100\% \tag{6}$$

$$TIC = \left(\sqrt{\frac{1}{n}\sum_{t=1}^{n}(\hat{y}_t - y_t)^2}\right) \Big/ \left(\sqrt{\frac{1}{n}\sum_{t=1}^{n}y_t^2} + \sqrt{\frac{1}{n}\sum_{t=1}^{n}\hat{y}_t^2}\right) \tag{7}$$

where: y_t – actual price at time t, $\hat{y}_t$ – predicted price; n – number of predictions. The lower the values of these statistics, the better the forecasting accuracy. RMSE depends on the scale of the forecasted variable, while MAPE and TIC are insensitive to it. MAPE is expressed as a percentage, and TIC assumes values from 0 to 1, with 0 indicating the best fit of the forecasts to actual values.

3. Data

The study material consisted of quarterly time series of the nominal prices of pine, spruce, beech, birch, and alder roundwood between 2005Q1 and 2020Q4 obtained from the reports of the Polish State Forest Information System. The time series for spruce roundwood prices is presented in Fig. 1.

Seasonal price fluctuations were analyzed using the Census X-11 method, with the statistical significance being evaluated by means of the F-test implemented in Statistica package 13.1. The exogenous leading variable was the Construction Confidence Index (CCI) provided by the Central Statistical Office (2021), which displays a similar course and direction of changes as roundwood prices (Fig. 1). This indicator assumes values from the range of −100 to +100, with positive values signifying a good sentiment, and negative values – a poor sentiment. Thus, an increase or decrease in the indicator means an improvement or deterioration in business conditions from the standpoint of construction companies. The CCI was chosen as a leading indicator because its values from preceding periods (lagged by one to three quarters) were significantly correlated with current roundwood prices.

4. Results

4.1. Stationarity and seasonality of the analyzed time series

The ADF and KPSS unit root tests indicate that the selected time series of nominal prices of roundwood prices are nonstationary at the 1% probability level, except KPSS for beech roundwood (Table 1). The first differences of all price time series are stationary.

The prices of selected roundwood species exhibited significant seasonality, with the highest values in Q1 and the lowest in Q3 (Table 2). The strongest seasonal fluctuations were observed in spruce prices, with an amplitude of 4.5% ($p < 0.0001$, F = 18.4); on the other hand the weakest fluctuations occurred in pine and alder roundwood prices ($p < 0.01$, F = 4.6).

4.2. Identification of the model structure

The shape of spruce roundwood prices time series indicated nonstationary of the mean, which increased from 49.57 to 59.60 € m^{-3} in the analyzed period (Fig. 1). It was found that the ACF of spruce roundwood prices gradually diminished following the pattern of an attenuated sine wave (Fig. 2a). In turn, the PACF was significant for the first two and seventeenth lags (Fig. 2b). This indicates a mixed process with quarterly seasonal elements, but does not allow for a clear

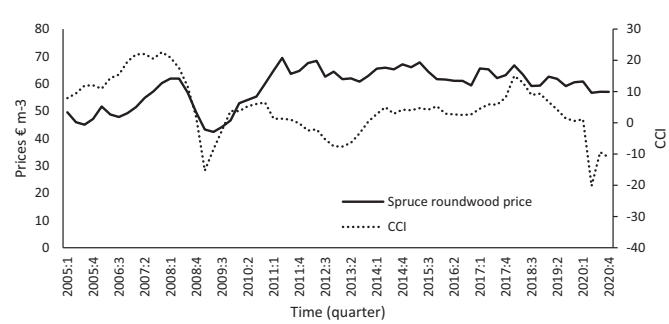

Fig. 1. Nominal prices of spruce roundwood and Construction Confidence Index (CCI) in Poland in 2005–2020.

Table 1
KPSS stationarity and ADF non-stationarity tests for the selected roundwood prices time series.

Series	Level		First differences	
	KPSS	ADF	KPSS	ADF
Pine	0.764[a]	0.806	0.041	−4.824[a]
Spruce	0.795[a]	0.362	0.092	−4.440[a]
Beech	0.310	−0.103	0.048	−5.931[a]
Birch	1.034[a]	0.178	0.032	−6.296[a]
Alder	0.552[a]	0.605	0.035	−4.980[a]

[a] Null hypothesis rejected at the 1% probability levels; the null hypothesis assumed time series stationarity in KPSS and nonstationarity in ADF.

Table 2
Seasonal variation of selected roundwood prices in Poland in 2005–2020.

Series	*F*-Statistic	*p*-Value	Amplitude[a] (%)
Pine	4.62	0.01	2.4
Spruce	18.35	0.0001	4.5
Beech	7.86	0.001	3.8
Birch	4.60	0.01	2.1
Alder	4.58	0.01	1.7

[a] Amplitude between Q1 and Q3, average values for the period 2005–2020.

identification of the structure and order of the model.

The parameters of the selected optimal models are given in Table 3; all of them are statistically significant. SARIMAX model was optimal for all series. The constant term reached statistical significance (at $p < 0.01$) only in the alder price model, and that why it was excluded from the other models. Residual analysis of the models revealed a roughly normal distribution of the residues. In addition, ACF and PACF analysis of the residuals indicated that they were not intercorrelated.

The developed models were used for short-term forecasting with time horizons of one (2021 Q1) to three (2021 Q3) quarters ahead (Table 4.)

A comparison of forecast prices for 2021Q1 with the actual prices available when this analysis was completed revealed that differences ranging between 0.47 € (beech) and + 0.54 € (birch). For 2021 Q2 and 2021Q3, the actual prices were not available. Accuracy analysis for two and three horizon forecasts is provided in following paragraph.

4.3. Forecast accuracy

Ex post forecasts of spruce prices were performed to evaluate the forecast accuracy of ARIMA, SARIMA and SARIMAX models. Forecasts were made in a dynamic way, assuming that model parameters should be re-estimated using new information known at the time of prediction. The first three forecasts were made for 2019Q1 with time horizons ranging from one quarter (based on data available in 2018Q4) to three quarters (based on data available in 2018Q2). Further successive forecasts were made up to 2020Q4 by adding consecutive quarters to the initial time series of timber prices and CCI values, and estimating the coefficients of the updated model.

Table 5 provides the actual prices of spruce roundwood as well as the predicted prices. For example, in 2019Q1 the real price was 62.60 € m^{-3}, while the prices forecasted by the ARIMA, SARIMA, and SARIMAX models in a one-quarter horizon (based on the data that were available as of 2018Q4) were 62.17 (±2.66), 62.45 (±2.85), and 61.21 (±2.43) € m^{-3}, respectively. The ARIMA, SARIMA, and SARIMAX forecasts made for the same period (2018Q1) based on the data available three quarters previously (as of 2018Q2) amounted to 62.73 (±6.54), 62.90 (±5.51), 61.40 (±3.76) € m^{-3}, respectively.

Forecast accuracy depended on the time horizon and the method applied. In a one-quarter horizon, accuracy measured by the standard error of forecast was similar for the three models (on average 2.6, 2.7, and 2.4 € m^{-3} for ARIMA, SARIMA, and SARIMAX, respectively). In longer time horizons, the differences in forecast accuracy increased; in a horizon of three quarters the mean standard error of SARIMAX forecasts was near half of that of ARIMA forecasts (3.6 € m^{-3} vs. 6.6 € m^{-3}).

The *ex post* measures of forecast accuracy, based on comparisons of predicted and actual prices, were as follows. In a one-quarter horizon, RMSR was 1.56, 1.43, and 1.09 € m^{-3}, and MAPE was 2.05, 1.90, and 1.46% for ARIMA, SARIMA, and SARIMAX, respectively. Forecast errors gradually increased with time horizon length; in the case of forecasts

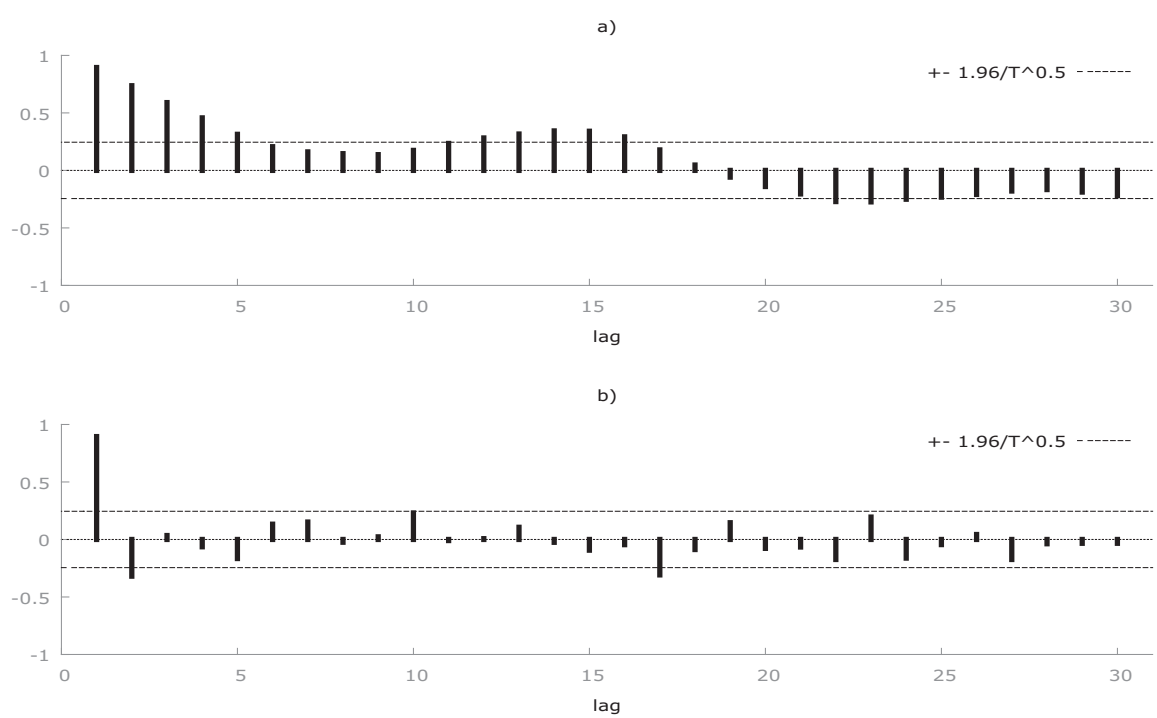

Fig. 2. Autocorrelation (ACF) (a), and partial autocorrelation (PACF) (b) functions of the spruce roundwood prices.

Table 3
Estimated models for selected roundwood prices.

Series	Estimated models		Goodness of fit		Residues	
			BIC	AIC	χ^2	Q_{LB}
Pine	SARIMAX (2,0,1)(0,1,1)^4	$(1–1.580\ L + 0.757\ L^2)(1\text{-}L^4)y_t = (1–0.468\ L)(1–0.549L^4)\ e_t + 0.218\ xc$	289.1	274.4	4.55	17.70
Spruce	SARIMAX (2,0,1)(0,1,1)^4 xc	$(1–1.531\ L + 0.560\ L^2)(1\text{-}L^4)y_t = (1–0.401\ L)(1 + 0.999\ L^4)e_t + 0.263\ xc$	287.9	277.4	0.789	26.24
Beech	SARIMAX (1,0,1)(0,1,1)^4 xc	$(1–0.751\ L)(1\text{-}L^4)y_t = (1–0.439\ L)(1 + 0.802\ L^4)e_t + 0.266\ xc$	297.7	285.2	4.41	24.00
Birch	SARIMAX (1,0,1)(0,1,1)	$(1–0.892\ L)(1\text{-}L^4)y_t = (1 + 0.208\ L)(1–0.912L^4)e_t + 0.147x_c$	298,1	287.6	5.60	28.29
Alder	SARIMAX (2,0,1)(0,1,1)^4	$(1–1.664\ L + 0.841\ L^2)(1\text{-}L^4)y_t = 0.522 + (1–0.999\ L)(1–0.998L^4)e_t + 0.109\ xc$	308.2	295.6	3.76	28.04

BIC – Bayesian information criterion, AIC – Akaike information criterion, x_c – Construction Confidence Index, Q_{lB} – Ljung-Box Q-statistic.

Table 4
Forecast of prices of selected roundwood.

	Time	Actual price	Price prediction	Lower 95%	Upper 95%
			Forecast		
Pine	2021Q1	56.49	56.39	51.39	61.38
	2021Q2	na	56.72	49.48	63.96
	2021Q3	na	56.56	48.35	64.18
Spruce	2021Q1	59.61	59.60	55.01	64.19
	2021Q2	na	58.76	50.54	66.99
	2021Q3	na	57.52	47.32	67.71
Beech	2021Q1	51.60	51.13	46.91	55.34
	2021Q2	na	48.26	41.77	54.75
	2021Q3	na	46.88	39.31	54.45
Birch	2021Q1	47.39	47.93	43.42	52.45
	2021Q2	na	47,53	40,92	54,12
	2021Q3	na	46.35	39,29	53.40
Alder	2021Q1	48.03	48.11	44.60	51.62
	2021Q2	na	49.32	44.97	53.66
	2021Q3	na	49.47	44.99	53.96

na – not available.

three quarters ahead RMSR was 2.27, 1.87, and 1.29 € m^{-3}, and MAPE was 3.18, 2.45, and 1.85% for ARIMA, SARIMA, and SARIMAX, respectively.

5. Discussion and conclusions

In the present work, timber prices were forecasted using three types of time series models: (1) a classical univariate autoregressive moving average model (ARIMA), (2) a univariate model with seasonal effects (SARIMA), and (3) a bivariate seasonal model with an exogenous variable (SARIMAX).

While time series models are widely used for forecasting in the forest sector, Leskinen and Kangas (2001) aptly observed that "one difficulty when using past time series data in forecasting future timber price development is the possibility of changes in the markets and in society at large." Therefore, in their predictions of timber prices those authors applied time series analysis (as a basic approach) in conjunction with expert judgments to discern future price trends and the probability of price peaks. Gejdoš et al. (2019), who used ARIMA as well as linear and multivariate exponential smoothing models for predicting the prices of beech sawlogs in Central Europe, found that six-month ARIMA forecasts were closest to the actual values, and so they recommended that model as a guide for decision making processes in management and trade. The present work shows that in a one-quarter time horizon, the examined models produced similar price projections. However, in the case of longer horizons, the SARIMA and SARIMAX models were found to be superior. While in ARIMA the forecasted prices stabilize at a certain level in successive quarters, SARIMA incorporates seasonal variation, while SARIMAX additionally reflects the effects of an exogenous variable (in this case changes in the construction market sentiment) that is external with respect to the data contained within the historical price time series.

Timber prices usually display seasonality arising from different levels of demand and supply in different seasons (Leefers and Potter-Witter, 2006) and seasonal harvest restrictions (Kolis et al., 2014). According to Michinaka et al. (2016) seasonal price fluctuations are mostly attributable to the effects of pests damaging timber quality. Sawmills

Table 5
Actual and forecasted prices of spruce roundwood in forecast horizons of one to three quarters ahead in 2019–2020 (€ m^{-3}).

Year quarter	Actual prices	Forecasted prices								
		ARIMA			SARIMA			SARIMAX		
		Q1	Q2	Q3	Q1	Q2	Q3	Q1	Q2	Q3
2019 Q1	62.60	62.17	62.48	62.73	62.45	61.64	62.90	61.21	61.50	61.40
		(2.66)	(4.42)	(6.54)	(2.85)	(4.44)	(5.51)	(2.43)	(3.17)	(3.67)
2019 Q2	61.77	61.34	60.75	61.03	61.98	61.81	61.02	61.37	59.69	59.45
		(2.63)	(4.38)	(6.56)	(2.82)	(4.39)	(5.49)	(2.43)	(3.24)	(3.67)
2019 Q3	59.14	60.60	60.01	59.48	59.43	59.67	59.51	58.82	58.41	57.61
		(2.61)	(4.35)	(6.46)	(2.80)	(4.35)	(5.43)	(2.41)	(3.23)	(3.63)
2019 Q4	60.55	60.08	62.22	61.67	59.83	60.18	60.41	59.06	58.71	58.30
		(2.63)	(4.31)	(6.52)	(2.77)	(4.31)	(5.39)	(2.39)	(3.21)	(3.61)
2020 Q1	60.82	60.86	60.04	62.38	62.38	63.51	63.86	62.94	61.26	60.90
		(2.57)	(4.52)	(6.57)	(2.75)	(4.27)	(5.33)	(2.38)	(3.18)	(3.58)
2020 Q2	56.66	60.12	60.17	59.25	58.99	58.80	58.99	55.50	54.73	54.93
		(2.55)	(4.25)	(6.49)	(2.72)	(4.24)	(5.29)	(2.37)	(3.16)	(3.56)
2020 Q3	57.05	55.61	60.39	60.44	54.61	57.57	60.20	57.63	55.57	56.48
		(2.54)	(4.22)	(6.52)	(2.76)	(4.24)	(5.24)	(2.38)	(3.14)	(3.54)
2020 Q4	57.03	58.00	55.94	60.97	59.01	54.91	58.49	56.72	57.39	55.54
		(2.55)	(4.20)	(6.47)	(2.77)	(4.29)	(5.31)	(2.36)	(3.19)	(3.59)
ME		2.59	4.32	6.55	2.68	4.31	5.37	2.39	3.19	3.60
RMSR		1.56	1.84	2.27	1.43	1.66	1.81	1.09	1.33	1.29
MAPE		2.05	2.49	3.18	1.90	2.26	2.45	1.46	1.93	1.85
TIC		0.013	0.031	0.038	0.024	0.028	0.030	0.018	0.022	0.025

Standard errors of forecast are given in brackets; RMSR – root mean square error; MAPE – mean absolute percent error; TIC – Theil's inequality coefficient.

strive to adjust their stocks of timber and reduce the acquisition of new timber in the spring and summer to avoid or lessen pest damage. Therefore, roundwood prices are low in the summer and start to increase in the autumn. In the present work, spruce roundwood prices were characterized by a much higher seasonal amplitude as compared to pine (4.5% vs. 2.4%).

ARIMA is not suitable for modeling time series with seasonality. The implementation of autocorrelation with remote lags of the modeled variable would require a very large number of parameters (high order p and q), resulting in overparameterization. As a result, this work employed SARIMA for forecasting timber prices with seasonal effects, although to date such models have not been widely used in the forest sector (timber price seasonality has been usually reflected in econometric models as a dummy variable (Brown et al., 2012)). Michinaka et al. (2016) used ARIMA models and an exponential smoothing method (ETS) to forecast the monthly prices of domestic roundwood of three of the most important species in Japan. For species with price seasonal effects, seasonality was incorporated into ARIMA by adding a seasonal component to the model, similarly as in SARIMA. It was found that in most cases ARIMA gave better results as compared to ETS. The best roundwood price model for tree species not characterized by price seasonality was ARIMA(0,1,0) with no autoregressive, moving average or seasonal terms (a random walk model). Malaty et al. (2007) who used time-series models including ARIMA, vector autoregression (VAR), and structural time series models (STSM) for forecasting Finnish pine sawlog prices concluded that in real-life forecasting situations it is rather difficult to obtain precise stumpage price estimates using solely a time series approach. Buongiorno et al. (1984) made forecasts of US softwood timber imports using a structural econometric model based on housing starts, prices, and past imports; a univariate time-series model using past imports only; and a bivariate time-series model of imports and housing starts, which was found to be the most useful. While being at least as accurate as the econometric model in one- and three-month forecasts, the bivariate model required much less information.

Models incorporating exogenous variables can generate good forecasts only if the lag of those variables (predictors) is large enough for them to be available already at the time of forecasting. The leading variable in this work was the Construction Confidence Index, whose values, lagged by one to three quarters, are correlated with timber prices because of the high demand for roundwood in the final stages of house construction (roofing and interior carpentry). Thus, the CCI provides advance information about increases or decreases in demand for roundwood, and indirectly about the direction of price changes.

Prices are usually affected by supply and demand. We tested the time series of roundwood supply by the State Forests as a potential exogenous variable (using different lags) but it turned out nonsignificant. This indicates that supply should be analyzed at a broader level. Further research on timber markets cointegration at the international level, especially between neighboring countries, is needed. Such cointegration and causality were tested by Yin et al. (2002) and Yin and Xu (2003) for selected US timber markets, with the obtained results having significant implication for price forecasting.

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Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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